

Onsite Payments

Receive payments without redirecting customers away from your website.

Recurring Billing

Enable subscription and tokenization payments.

[Subscriptions](#)

[Tokenization](#)

[Additional methods of integrating](#)

Pay Now Button

Generate Pay Now HTML button code.

Shopping Carts

Choose from one of our many plugins.

Custom Payment Integration

Build a checkout form and receive payments securely from our payment platform. This process can be used for both one-time and recurring payments.

Live: <https://www.payfast.co.za/eng/process>

Sandbox: <https://sandbox.payfast.co.za/eng/process>

Quick start: It is recommended that you create your own [Sandbox](#) account to test your integration.

Simple Form Integration

```
<form action="https://www.payfast.co.za/eng/process" method="post">
```

```
  <input type="hidden" name="merchant_id" value="10000100">
```

```
  <input type="hidden" name="merchant_key" value="46f0cd694581a">
```

```
  <input type="hidden" name="amount" value="100.00">
```

```
<input type="hidden" name="item_name" value="Test Product">

<input type="submit">

</form>
```

Step 1: Create your checkout form

Create a checkout page with an HTML form, usually hidden, with a number of fields containing all the necessary information needed for Payfast to process the payment. Most shopping cart platforms will have the customer click on a 'Confirm Order' or 'Pay Now' button to submit the form and be redirected to the Payfast Payment page.

Merchant details

merchant_idinteger, 8 char**REQUIRED**

The Merchant ID as given by the Payfast system. Used to uniquely identify the receiving account. This can be found on the merchant's settings page.

merchant_keystring**REQUIRED**

The Merchant Key as given by the Payfast system. Used to uniquely identify the receiving account. This provides an extra level of certainty concerning the correct account as both the ID and the Key must be correct in order for the transaction to proceed. This can be found on the merchant's settings page.

return_urlstring, no char limit**OPTIONAL**

The URL where the user is returned to after payment has been successfully taken.

cancel_urlstring, no char limit**OPTIONAL**

The URL where the user should be redirected should they choose to cancel their payment while on the Payfast system.

notify_urlstring, no char limit**OPTIONAL**

The URL which is used by Payfast to post the [Instant Transaction Notifications](#) (ITNs) for this transaction.

fica_idnumberinteger, 13 char**OPTIONAL**

The Fica ID Number provided of the buyer must be a valid South African ID Number.

For the notify_url mentioned, a variable can be specified globally on the Merchant's Payfast account or overridden on a per transaction basis. The value provided during a transaction overrides the global setting.

Merchant Details

<input type="hidden" name="merchant_id" value="10000100">

<input type="hidden" name="merchant_key" value="46f0cd694581a">

<input type="hidden" name="return_url" value="https://www.example.com/success">

<input type="hidden" name="cancel_url" value="https://www.example.com/cancel">

<input type="hidden" name="notify_url" value="https://www.example.com/notify">

Customer details

name_firststring, 100 char **OPTIONAL**

The customer's first name.

name_laststring, 100 char **OPTIONAL**

The customer's last name.

email_addresstring, 100 char **OPTIONAL**

The customer's email address.

cell_numberstring, 100 char **OPTIONAL**

The customer's valid cell number. If the email_address field is empty, and cell_number provided, the system will use the cell_number as the username and auto login the user, if they do not have a registered account

Customer Details

<input type="hidden" name="name_first" value="John">

<input type="hidden" name="name_last" value="Doe">

<input type="hidden" name="email_address" value="john@doe.com">

<input type="hidden" name="cell_number" value="0823456789">

Transaction details

m_payment_idstring, 100 char **OPTIONAL**

Unique payment ID on the merchant's system.

amountdecimal**REQUIRED**

The amount which the payer must pay in ZAR.

item_namestring, 100 char **REQUIRED**

The name of the item being charged for, or in the case of multiple items the order number.

item_descriptionstring, 255 char **OPTIONAL**

The description of the item being charged for, or in the case of multiple items the order description.

custom_int<1..5>integer, 255 char**OPTIONAL**

A series of 5 custom integer variables (*custom_int1*, *custom_int2*...) which can be used by the merchant as pass-through variables. They will be posted back to the merchant at the completion of the transaction.

custom_str<1..5>string, 255 char **OPTIONAL**

A series of 5 custom string variables (*custom_str1*, *custom_str2*...) which can be used by the merchant as pass-through variables. They will be posted back to the merchant at the completion of the transaction.

Transaction Details

```
<input type="hidden" name="m_payment_id" value="01AB">
```

```
<input type="hidden" name="amount" value="100.00">
```

```
<input type="hidden" name="item_name" value="Test Item">
```

```
<input type="hidden" name="item_description" value="A test product">
```

```
<input type="hidden" name="custom_int1" value="2">
```

```
<input type="hidden" name="custom_str1" value="Extra order information">
```

Transaction options

email_confirmationboolean, 1 char**OPTIONAL**

Whether to send an email confirmation to the merchant of the transaction. The email confirmation is automatically sent to the payer. 1 = on, 0 = off

confirmation_addresstring, 100 char**OPTIONAL**

The email address to send the confirmation email to. This value can be set globally on your account. Using this field will override the value set in your account for this transaction.

Transaction Options

```
<input type="hidden" name="email_confirmation" value="1">
```

<input type="hidden" name="confirmation_address" value="john@doe.com">

Payment methods

payment_methodstring, 3 char **OPTIONAL**

When this field is set, only the SINGLE payment method specified can be used when the customer reaches Payfast. If this field is blank, or not included, then all available payment methods will be shown.

The values are as follows:

- 'ef' – EFT
- 'cc' – Credit card
- 'dc' – Debit card
- 'mp' – Masterpass Scan to Pay
- 'mc' – Mobicred
- 'sc' – SCode
- 'ss' – SnapScan
- 'zp' – Zapper
- 'mt' – MoreTyme
- 'rc' – Store card
- 'mu' – Mukuru
- 'ap' – Apple Pay
- 'sp' – Samsung Pay
- 'cp' – Capitec Pay
- 'ab' – Absa Pay
- 'gp' – Google Pay
- 'nd' – Nedbank Direct EFT
- 'pf' – Payflex. Buy Now, Pay Later

Payment Methods

<input type="hidden" name="payment_method" value="cc">

Recurring Billing and Split Payment form options

([See Recurring Billing subscriptions and Split Payments for additional fields](#))

Step 2: Create security signature

A generated MD5 signature must be passed as an additional hidden input and this hash is then used to ensure the integrity of the data transfer.

Security Signature

```
<input type="hidden" name="signature"
value="f103e22c0418655fb03991538c51bfd5">
```

To generate the signature:

1. Concatenation of the name value pairs of all the non-blank variables with ‘&’ used as a separator

Variable order: The pairs must be listed in the **order in which they appear** in the attributes description. eg. name_first=John&name_last=Doe&email_address=...

** Do not use the **API signature** format, which uses alphabetical ordering!*

2. Add your passphrase

The passphrase is an extra security feature, used as a ‘salt’, and is set by the Merchant in the Settings section of their Payfast Dashboard.

Add the passphrase to the end of the below string.

E.g. name_first=John&name_last=Doe&email_address=...&passphrase=...

The resultant URL encoding must be in upper case (eg. http%3A%2F%2F), and spaces encoded as ‘+’.

3. MD5 the parameter string and pass it as a hidden input named “signature”

Troubleshooting: For troubleshooting signature mismatch issues check out our knowledge base [Common causes of a failed integration / signature mismatch](#)

Signature generation

Select library

```
<?php
```

```
/**
```

```

* @param array $data
* @param null $passPhrase
* @return string
*/

function generateSignature($data, $passPhrase = null) {
    // Create parameter string
    $pfOutput = "";
    foreach( $data as $key => $val ) {
        if($val !== "") {
            $pfOutput .= $key .'=' . urlencode( trim( $val ) ) . '&';
        }
    }
    // Remove last ampersand
    $getString = substr( $pfOutput, 0, -1 );
    if( $passPhrase !== null ) {
        $getString .= '&passphrase=' . urlencode( trim( $passPhrase ) );
    }
    return md5( $getString );
}

```

Step 3: Send the customer to Payfast for payment

On submission of your form, your customers will be redirected to the secure Payfast payments page.

Your form should be setup to post to either Payfast's live URL or the Sandbox for testing.

Full form implementation

Select library

<?php

// Construct variables

\$cartTotal = 10.00; // This amount needs to be sourced from your application

\$passphrase = 'jt7NOE43FZPn';

```

$data = array(

    // Merchant details

    'merchant_id' => '10000100',

    'merchant_key' => '46f0cd694581a',

    'return_url' => 'http://www.yourdomain.co.za/return.php',

    'cancel_url' => 'http://www.yourdomain.co.za/cancel.php',

    'notify_url' => 'http://www.yourdomain.co.za/notify.php',

    // Buyer details

    'name_first' => 'First Name',

    'name_last' => 'Last Name',

    'email_address'=> 'test@test.com',

    // Transaction details

    'm_payment_id' => '1234', //Unique payment ID to pass through to notify_url

    'amount' => number_format( sprintf( '%.2f', $cartTotal ), 2, '.', '' ),

    'item_name' => 'Order#123'

);

$signature = generateSignature($data, $passphrase);

$data['signature'] = $signature;

// If in testing mode make use of either sandbox.payfast.co.za or www.payfast.co.za

$testingMode = true;

$pfHost = $testingMode ? 'sandbox.payfast.co.za' : 'www.payfast.co.za';

$htmlForm = '<form action="https://'.$pfHost.'/eng/process" method="post">';

foreach($data as $name=> $value)

{

    $htmlForm .= '<input name="'.$name.'" type="hidden" value="'.$value.'" />';

}

```



```
$htmlForm .= '<input type="submit" value="Pay Now" /></form>';  
echo $htmlForm;
```

Step 4: Confirm payment is successful

Before the customer is returned to the URL you specified in the “return_url” parameter, Payfast will send a notification to your “notify_url” page.

4.1. Setting up your notification page:

On receiving the payment notification from Payfast, return a header 200 to prevent further retries.

If no 200 response is returned the notification will be re-sent immediately, then after 10 minutes and then at exponentially longer intervals until eventually stopping.

If you are testing locally you will need to have a publicly accessible URL (notify_url) in order to receive the notifications, consider using tools such as [NGROK](#) or [Expose](#) to expose your local development server to the Internet.

4.2. Receive the notification parameters

As part of the notification you can expect to receive the following parameters:

Transaction details

m_payment_idstring, 100 char**OPTIONAL**

Unique payment ID on the merchant’s system.

pf_payment_idinteger**REQUIRED**

Unique transaction ID on Payfast.

payment_statusstring (CANCELLED / COMPLETE)**REQUIRED**

After a successful payment the status sent will be COMPLETE. When a subscription is cancelled the status will be CANCELLED.

item_namestring, 100 char**REQUIRED**

The name of the item being charged for, or in the case of multiple items the order number.

item_descriptionstring, 255 char**OPTIONAL**

The description of the item being charged for, or in the case of multiple items the order description.

amount_grossdecimal**OPTIONAL**

The total amount the customer paid in ZAR.

amount_feedecimal**OPTIONAL**

The total in fees which was deducted from the amount in ZAR.

amount_netdecimal**OPTIONAL**

The net amount credited to the merchant's account in ZAR.

custom_int<1..5>integer, 255 char**OPTIONAL**

The series of 5 custom integer variables (custom_int1, custom_int2...) originally passed by the receiver during the payment request.

custom_str<1..5>string, 255 char**OPTIONAL**

The series of 5 custom string variables (custom_str1, custom_str2...) originally passed by the receiver during the payment request.

Customer details

name_firststring, 100 char**OPTIONAL**

The customer's first name.

name_laststring, 100 char**OPTIONAL**

The customer's last name.

email_addresstring, 100 char**OPTIONAL**

The customer's email address.

Merchant details

merchant_idinteger, 8 char**REQUIRED**

The Merchant ID as given by the Payfast system. Used to uniquely identify the receiver's account.

Recurring Billing details

tokenstring, 36 char**REQUIRED**

Unique ID on Payfast that represents the subscription.

billing_date, YYYY-MM-DD**OPTIONAL**

The date from which future subscription payments will be made. Eg. 2020-01-01.

Defaults to current date if not set.

Security information

signature MD5 hash in lower case**OPTIONAL**

A security signature of the transmitted data taking the form of an MD5 hash of all the url encoded submitted variables.

Example of a Transaction Webhook Payload

Select library

```
$ITN_Payload = [  
  'm_payment_id' => 'SuperUnique1',  
  'pf_payment_id' => '1089250',  
  'payment_status' => 'COMPLETE',  
  'item_name' => 'test+product',  
  'item_description' => ,  
  'amount_gross' => 200.00,  
  'amount_fee' => -4.60,  
  'amount_net' => 195.40,  
  'custom_str1' => ,  
  'custom_str2' => ,  
  'custom_str3' => ,  
  'custom_str4' => ,  
  'custom_str5' => ,  
  'custom_int1' => ,  
  'custom_int2' => ,  
  'custom_int3' => ,  
  'custom_int4' => ,
```

```
'custom_int5' => ,  
'name_first' => ,  
'name_last' => ,  
'email_address' => ,  
'merchant_id' => '10012577',  
'signature' => 'ad8e7685c9522c24365d7ccea8cb3db7'  
];
```

4.3. Conduct security checks

Conduct four security checks to ensure that the data you are receiving is correct, from the correct source and hasn't been altered; you should not continue the process if a test fails!

Notification page

Select library

<?php

```
// Tell Payfast that this page is reachable by triggering a header 200
```

```
header( 'HTTP/1.0 200 OK' );
```

```
flush();
```

```
define( 'SANDBOX_MODE', true );
```

```
$pfHost = SANDBOX_MODE ? 'sandbox.payfast.co.za' : 'www.payfast.co.za';
```

```
// Posted variables from ITN
```

```
$pfData = $_POST;
```

```
// Strip any slashes in data
```

```
foreach( $pfData as $key => $val ) {
```

```
    $pfData[$key] = stripslashes( $val );
```

```
}
```

```
// Convert posted variables to a string
foreach( $pfData as $key => $val ) {
    if( $key !== 'signature' ) {
        $pfParamString .= $key .'=' . urlencode( $val ) . '&';
    } else {
        break;
    }
}
}
```

```
$pfParamString = substr( $pfParamString, 0, -1 );
```

- **Verify the signature**

Verify the security signature in the notification; this is done in a similar way that the signature that you generated for stage one of the user payment flow.

The string that gets created needs to include all fields posted from Payfast.

Verify Signature

Select library

<?php

```
function pfValidSignature( $pfData, $pfParamString, $pfPassphrase = null ) {
    // Calculate security signature
    if($pfPassphrase === null) {
        $tempParamString = $pfParamString;
    } else {
        $tempParamString = $pfParamString.'&passphrase=' . urlencode( $pfPassphrase );
    }

    $signature = md5( $tempParamString );
```

```
return ( $pfData['signature'] === $signature );  
}
```

- **Check that the notification has come from a valid Payfast domain**

The following is a list of valid domains:

- www.payfast.co.za
- w1w.payfast.co.za
- w2w.payfast.co.za
- sandbox.payfast.co.za

Check valid Payfast domain

Select library

<?php

```
function pfValidIP() {  
    // Variable initialization  
  
    $validHosts = array(  
        'www.payfast.co.za',  
        'sandbox.payfast.co.za',  
        'w1w.payfast.co.za',  
        'w2w.payfast.co.za',  
    );  
  
    $validIps = [];  
  
    foreach( $validHosts as $pfHostname ) {  
        $ips = gethostbyname( $pfHostname );  
  
        if( $ips !== false )  
            $validIps = array_merge( $validIps, $ips );  
    }  
}
```

```

}

// Remove duplicates
$validIps = array_unique( $validIps );
$referrerIp = gethostbyname(parse_url($_SERVER['HTTP_REFERER'])['host']);
if( in_array( $referrerIp, $validIps, true ) ) {
    return true;
}
return false;
}

```

- **Compare payment data**

The amount you expected the customer to pay should match the “amount_gross” value sent in the notification.

Compare payment data

Select library

<?php

```

function pfValidPaymentData( $cartTotal, $pfData ) {
    return !(abs((float)$cartTotal - (float)$pfData['amount_gross']) > 0.01);
}

```

- **Perform a server request to confirm the details**

Validate the data that you have received from Payfast by contacting our server and confirming the order details.

Live: <https://www.payfast.co.za/eng/query/validate>

Sandbox: <https://sandbox.payfast.co.za/eng/query/validate>

Curl Options

CURLOPT_RETURNTRANSFER	Set to TRUE to return the transfer as a string of the return value of curl_exec() instead of outputting it out directly.
CURLOPT_HEADER	Set to FALSE to exclude the header in the output.
CURLOPT_SSL_VERIFYHOST	Set to 2 to check the existence of a common name in the SSL peer certificate, and to also verify that it matches the hostname provided.
CURLOPT_SSL_VERIFYPEER	This option determines whether curl verifies the authenticity of the peer's certificate. The default value is 1.
CURLOPT_URL	Set to the Payfast query validation URL, eg. https://www.payfast.co.za/eng/query/validate . This can also be set when initializing a session with curl_init().
CURLOPT_POST	Set to TRUE to do a regular HTTP POST. This POST is the normal application/x-www-form-urlencoded kind, most commonly used by HTML forms.
CURLOPT_POSTFIELDS	The full data to post via HTTP to Payfast. This parameter is passed as a urlencoded string, such as: m_payment_id=01&pf_payment_id=01234..

Server request to confirm details

Select library

<?php

```
function pfValidServerConfirmation( $pfParamString, $pfHost = 'sandbox.payfast.co.za',  
$pfProxy = null ) {  
  
    // Use cURL (if available)  
  
    if( in_array( 'curl', get_loaded_extensions(), true ) ) {  
  
        // Variable initialization  
  
        $url = 'https://'. $pfHost .'/eng/query/validate';  
  
  
  
  
        // Create default cURL object  
  
        $ch = curl_init();
```



```

// Set cURL options - Use curl_setopt for greater PHP compatibility

// Base settings

curl_setopt( $ch, CURLOPT_USERAGENT, NULL ); // Set user agent

curl_setopt( $ch, CURLOPT_RETURNTRANSFER, true ); // Return output as string
rather than outputting it

curl_setopt( $ch, CURLOPT_HEADER, false ); // Don't include header in output

curl_setopt( $ch, CURLOPT_SSL_VERIFYHOST, 2 );

curl_setopt( $ch, CURLOPT_SSL_VERIFYPEER, true );


// Standard settings

curl_setopt( $ch, CURLOPT_URL, $url );

curl_setopt( $ch, CURLOPT_POST, true );

curl_setopt( $ch, CURLOPT_POSTFIELDS, $pfParamString );

if( !empty( $pfProxy ) )

    curl_setopt( $ch, CURLOPT_PROXY, $pfProxy );


// Execute cURL

$response = curl_exec( $ch );

curl_close( $ch );

if ( $response === 'VALID' ) {

    return true;

}

}

return false;

}

```

Bringing the checks together

Select library

<?php

\$check1 = pfValidSignature(\$pfData, \$pfParamString);

\$check2 = pfValidIP();

\$check3 = pfValidPaymentData(\$cartTotal, \$pfData);

\$check4 = pfValidServerConfirmation(\$pfParamString, \$pfHost);

if(\$check1 && \$check2 && \$check3 && \$check4) {

 // All checks have passed, the payment is successful

} else {

 // Some checks have failed, check payment manually and log for investigation

}